

PINN and KANN approaches for shell structures: sampling strategies and experimental analysis

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Abstract: This study compared standard Physics-Informed Neural Networks (PINNs) and Kolmogorov–Arnold Networks (KANs) for thin-shell problems approaching the Kirchhoff–Love limit. With $t=0.01$, PINNs (3×50 neurons) achieved faster, more stable convergence and lower L^2 errors than KANs (3×5 neurons), despite having more parameters and halving training time. KANs’ slower convergence and higher final errors indicate limited representational capacity in the current setup. Sampling affected results: Hammersley points yielded slightly better accuracy, particularly for PINNs. Overall, PINNs proved more robust and computationally efficient, while KANs, though promising for compact parametrization, need refinement.

Keywords: shell structures, PINN, Kolmogorov–Arnold Networks.

1 Introduction

Partial differential equations (PDEs) describe a wide range of physical phenomena in engineering and science. While exact solutions exist for only a few simple cases, most practical problems with complex geometries, boundary conditions, or material behavior require numerical methods such as the finite element method (FEM). FEM has been highly successful and is widely used in both commercial and open-source frameworks, but its application to thin shell structures can be challenging. In shell modeling, issues such as numerical locking and the need for carefully designed shape functions often demand user expertise and significant fine-tuning, as noted by Bastek and Kochmann [1].

In recent years, deep learning has emerged as an alternative or complement to classical PDE solvers. Among these methods, Physics-Informed Neural Networks (PINNs), introduced by Raissi et al. [2], have gained considerable attention. Unlike purely data-driven models, PINNs embed the governing equations directly into the loss function, enabling the network to approximate solutions without requiring training data. PINNs have been successfully applied in diverse areas, including fluid dynamics, heat transfer, and solid mechanics, and have shown promising results for structural problems such as plates and shells.

However, most existing PINNs rely on multilayer perceptrons (MLPs). While MLPs are universal function approximators, they may require a large number of parameters, suffer from spectral bias, and lack interpretability, as discussed by Wu et al. [3]. To address these limitations, the Kolmogorov–Arnold Network (KAN), proposed by Liu et al. [4], replaces fixed activation functions with learnable ones, often using B-splines or other basis functions with strong approximation properties. This architecture can reduce parameter counts, improve numerical efficiency, and better align with classical approximation theory. Integrating KAN into PINNs—sometimes referred to as KINN or PIKAN—offers a promising alternative for solving PDEs in various formulations, including the energy form commonly used in structural mechanics.

Motivated by these developments, this work applies both standard PINN and KAN-based PINN frameworks to the mechanical analysis of shell structures, aiming to evaluate their accuracy, convergence behavior, and robustness compared to FEM solutions. The following section presents the theoretical background of shell models, focusing on the formulations adopted in this study.

2 Shell Theory short background

In this work, the mechanical behavior of shells is described using the linear five-parameter Naghdi [5] shell model in terms of differential geometry Ciarlet [6]. A shell is represented by its midsurface $\boldsymbol{\phi}(\xi_1, \xi_2)$, which maps the reference domain $\omega \subset \mathbb{R}^2$ into the physical space $\Omega \subset \mathbb{R}^3$. At each point, a covariant basis $\mathbf{a}_\alpha = \partial \boldsymbol{\phi} / \partial \xi_\alpha$, ($\alpha=1,2$) spans the tangent plane, and the unit normal $\mathbf{a}_3$ is obtained from their cross product. The in-plane geometry is characterized by the metric tensor $a_{\alpha\beta} = \mathbf{a}_\alpha \cdot \mathbf{a}_\beta$ while the surface element is given by $dS = \sqrt{ad\xi_1 d\xi_2}$.

The displacement field $\mathbf{U}(\xi_1, \xi_2, \xi_3)$ is expressed as $\mathbf{U} = \mathbf{u}(\xi_1, \xi_2) + \xi_3 \theta_\lambda(\xi_1, \xi_2) \mathbf{a}_\lambda$ where $\mathbf{u}$ are the midsurface displacements and θ_λ are the rotations of the normal fiber about the tangent directions $\mathbf{a}_\lambda$. This leads to three displacement components and two rotation components as degrees of freedom.

In the small-strain regime, the strain measures are:

Membrane strain

$$e_{\alpha\beta}(\mathbf{u}) = \frac{1}{2}(u_{\alpha|\beta} + u_{\beta|\alpha}) - b_{\alpha\beta}u_3, \quad (1)$$

Bending strain

$$\kappa_{\alpha\beta}(\mathbf{u}) = \frac{1}{2}(\theta_{\alpha|\beta} + \theta_{\beta|\alpha} - b_\beta^\lambda u_{\lambda|\alpha} - b_\alpha^\lambda u_{\lambda|\beta}) + c_{\alpha\beta}u_3, \quad (2)$$

Shear strain

$$\gamma_\alpha(\mathbf{u}, \boldsymbol{\theta}) = \theta_\alpha + u_{3,\alpha} + b_\alpha^\lambda u_\lambda. \quad (3)$$

where $b_{\alpha\beta}$ and $c_{\alpha\beta}$ are curvature-related coefficients from the second and third fundamental forms, respectively.

The total potential energy of a homogeneous isotropic shell is

$$\Pi[\mathbf{u}, \boldsymbol{\theta}] = \frac{1}{2} \int_\omega \left[\mathbb{C}^{\alpha\beta\sigma\rho} \left(t e_{\alpha\beta} e_{\sigma\rho} + \frac{t^3}{12} \kappa_{\alpha\beta} \kappa_{\sigma\rho} \right) + \mathbb{D}^{\alpha\beta} t \kappa \gamma_\alpha \gamma_\beta \right] dS - W_{ext}, \quad (4)$$

where t is the thickness, $\mathbb{C}^{\alpha\beta\sigma\rho}$ and $\mathbb{D}^{\alpha\beta}$ are the in-plane and shear elastic tensors, and $\kappa=5/6$ is the shear correction factor.

The equilibrium configuration follows from the principle of minimum potential energy:

$$\{\mathbf{u}^*, \boldsymbol{\theta}^*\} = \arg. \min. \{\Pi[\mathbf{u}, \boldsymbol{\theta}]\} \quad (5)$$

which serves as the basis for the weak form used in this work and directly aligns with the physics-informed neural network (PINN) formulation.

3 Methodology

This study investigates two deep learning frameworks for solving the small-strain Naghdi shell problem: a standard Physics-Informed Neural Network (PINN) and a Kolmogorov–Arnold Network (KAN)-based PINN, also referred to here as KAN-PINN. Both methods share the same physical formulation, based on the weak form of the shell equations described in Section 2, but differ in their internal neural architecture.

In the PINN framework, the displacement and rotation fields $(\mathbf{u}, \boldsymbol{\theta})$ are approximated by a multi-layer perceptron (MLP). The governing equations are embedded into the loss function by directly evaluating the strain measures $e_{\alpha\beta}, \kappa_{\alpha\beta}, \gamma_\alpha$ at collocation points and computing the corresponding energy contributions (membrane, bending, and shear) as in eq. (5). The minimization of this energy, minus the external work, enforces the physical laws without requiring training data.

The KAN-PINN uses the same physical loss definition but replaces the MLP with a Kolmogorov–Arnold Network, in which the activation functions are not fixed but learned during training. In this work, we use B-spline functions to parameterize these activations, following the original KAN proposal. This approach reduces the number of trainable parameters and aligns the network’s function approximation with classical numerical basis functions, potentially improving accuracy and convergence.

Both networks are trained using the same dataset of collocation points in the reference domain ω , generated with low-discrepancy Sobol sequences to ensure good spatial coverage. The weak form formulation is chosen over the strong form because it requires lower-order derivatives, incorporates natural boundary conditions directly, and

has shown better convergence properties in previous studies.

Fig. 1 illustrates the conceptual difference between the two frameworks. Both receive the same geometric mapping $\Phi(\xi_1, \xi_2)$, material parameters, and boundary conditions as input. In the PINN, the approximation is carried out by an MLP, while in the KAN-PINN, the approximation layer uses learnable B-spline-based activation functions. In both cases, the predicted fields are used to assemble the energy functional, which is minimized during training.

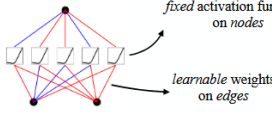
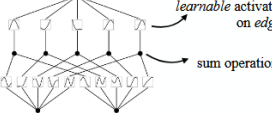
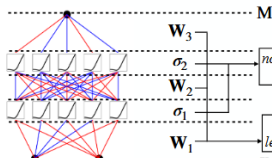
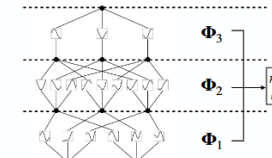
Model	Multi-Layer Perceptron (MLP)	Kolmogorov-Arnold Network (KAN)
Theorem	Universal Approximation Theorem	Kolmogorov-Arnold Representation Theorem
Formula (Shallow)	$f(\mathbf{x}) \approx \sum_{i=1}^{N(e)} a_i \sigma(\mathbf{w}_i \cdot \mathbf{x} + b_i)$	$f(\mathbf{x}) = \sum_{q=1}^{2n+1} \Phi_q \left(\sum_{p=1}^n \phi_{q,p}(x_p) \right)$
Model (Shallow)	(a)  fixed activation functions on nodes learnable weights on edges	(b)  learnable activation functions on edges sum operation on nodes
Formula (Deep)	$\text{MLP}(\mathbf{x}) = (\mathbf{W}_3 \circ \sigma_2 \circ \mathbf{W}_2 \circ \sigma_1 \circ \mathbf{W}_1)(\mathbf{x})$	$\text{KAN}(\mathbf{x}) = (\Phi_3 \circ \Phi_2 \circ \Phi_1)(\mathbf{x})$
Model (Deep)	(c)  $\mathbf{W}_3$ σ_2 $\mathbf{W}_2$ σ_1 $\mathbf{W}_1$ $\mathbf{x}$ nonlinear, fixed linear, learnable	(d)  Φ_3 Φ_2 Φ_1 $\mathbf{x}$ nonlinear, learnable

Figure 1: MLPs vs. KANs. Liu et al. [4].

3.1 Sampling strategies

Wu et al. [3] present a comprehensive analysis of sampling strategies for PINNs, covering both non-adaptive and residual-based adaptive methods. For non-adaptive approaches, low-discrepancy sequences such as Sobol and Hammersley are highlighted for their ability to distribute points almost uniformly over the domain, reducing the gaps and clusters often seen in purely random sampling.

The Sobol sequence is deterministically constructed to fill the space with high uniformity, improving domain coverage and reducing the variance of approximation errors. The Hammersley sequence follows the same low-discrepancy principle but uses a construction based on prime numbers and radical inverse functions, showing performance comparable to Sobol in many problems.

These methods are especially useful for problems with complex geometries and for PINNs formulated in the weak form, as they ensure that the evaluation of energy functionals is less sensitive to the exact location of the collocation points. While they do not capture local regions of high error (as adaptive methods do), they offer simplicity, reproducibility, and good initial domain coverage. This makes them a robust choice for reference experiments or comparative studies between architectures, such as the PINN vs. KAN case considered in this work.

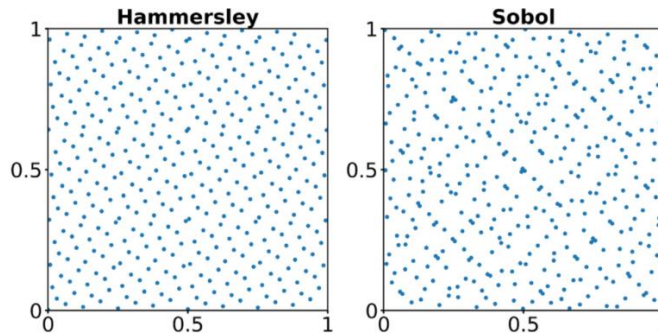


Figure 2: Examples of 400 points generated in $[0,1]^2$ using Hammersley and Sobol sampling methods. Wu et al. [3]

4 NUMERICAL TEST

The Scordelis–Lo roof benchmark (Fig. 3) is a widely used test in computational mechanics to evaluate shell formulations, particularly in the presence of membrane locking, as discussed by Ding et al. [7]. Locking occurs when the chosen approximation space cannot represent the actual deformation modes of the physical problem. This issue becomes critical for thin shells, where pure bending or membrane states dominate and require sufficient flexibility in the numerical method to avoid over-constraining the solution. Various improved finite element formulations have been developed to mitigate locking, such as those presented by Chapelle and Bathe [8], but they are often complex to implement and may rely on empirically tuned parameters, including the splitting parameter proposed by Hale et al. [9]. Despite decades of research, designing shell finite elements that are free from locking in both membrane- and bending-dominated regimes remains challenging, as emphasized again by Hale et al. [9].

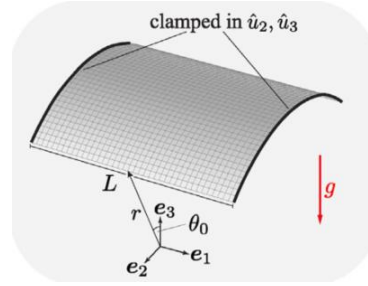


Figure 3: Scordelis-Lo roof consisting of a cylindrical shape and curved edges.

This benchmark is particularly relevant because it combines membrane-dominated regions with bending-dominated boundary layers, making it a stringent test for any numerical method. We hypothesize that Physics-Informed Neural Networks (PINNs), as introduced by Raissi et al. [2], due to their continuous function space and nonlinear approximation capabilities, can naturally overcome some of these degeneracies without the need for specialized locking-mitigation schemes. In this work, we extend the analysis to compare PINNs with Kolmogorov–Arnold Networks (KANs), proposed by Liu et al. [4], which replace standard MLP activations with learnable basis functions. This modification may enhance the spectral representation of the solution and accelerate convergence.

The problem setup follows the classical definition provided by Belytschko et al. [10], where the shell midsurface is parameterized by

$$\phi(\xi_1, \xi_2) = \left\{ \xi_1, \frac{1}{2} \sin(\xi_2), \frac{1}{2} \cos(\xi_2) \right\}, \quad (6)$$

with $\xi_1 \in [-0.5, 0.5] \in \xi_2 \in [-2\pi/9, 2\pi/9]$ corresponding to a cylindrical segment of radius $r=0.5$ and central angle $\theta_0 = 40^\circ$. The shell thickness-to-length ratio varies to test the method's robustness, and Poisson's ratio is set to $\nu=0$. Gravity loading is applied, and the ends of the cylinder are clamped in the $\hat{u}_2$ and $\hat{u}_3$ directions. Dirichlet boundary conditions are enforced through smooth trial functions to eliminate rigid body modes.

The main quantity of interest is the vertical displacement $\hat{u}_3$ at the midpoint of the free edge, compared to the reference solution $\hat{u}_3 = -0.3024$ for a thickness-to-length ratio of 0.05, Belytschko et al. [10]. Additionally, a reference FEM solution is obtained using selective reduced integration to mitigate locking.

For both PINNs and KANs, we adopt the weak form of the shell equilibrium equations, trained with $N_c=15000$ collocation points. The PINN uses a standard MLP architecture with the GELU activation function, while the KAN replaces the activations with B-spline-based learnable basis expansions Liu et al. [4]. Training is performed until convergence, and results are compared in terms of L_2 error against the FEM reference and displacement accuracy.

Both network architectures were based in 3 hidden layers, however, PINN had 50 neurons per layer while KAN had 5. The experiment was performed under the pre-defined number of 1000 epochs evaluation and the same L-BFGS optimizer was deployed.

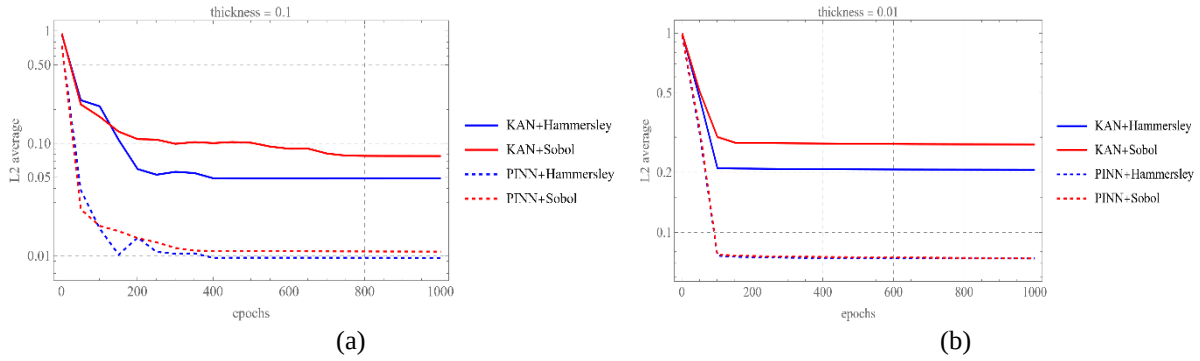


Figure 4: PINN and KAN performing against thickness variation from 0.1 to 0.01.

5 CONCLUSION

The Scordelis–Lo roof benchmark provided a stringent test by combining membrane-dominated regions with bending-dominated boundary layers, thus approaching the theoretical limit of Kirchhoff–Love shells when the thickness was reduced. Under these challenging conditions, both Physics-Informed Neural Networks (PINNs) and Kolmogorov–Arnold Networks (KANs) delivered physically consistent results without the need for explicit locking-mitigation techniques, validating their ability to operate in continuous function spaces.

When comparing sampling strategies, the Hammersley sequence generally led to slightly faster convergence than Sobol sampling, particularly in the initial training stages. However, for both methods, the asymptotic error levels were similar, suggesting that the sampling choice mainly affects early convergence rates rather than final accuracy.

From a computational perspective, PINNs required on average 1.5 hours of training, whereas KANs—despite having significantly fewer neurons per layer—took about 3 hours. This longer runtime is likely linked to the overhead introduced by learning basis functions, but it could be mitigated by optimized B-spline implementations or hybrid training schemes.

While KANs did not yet surpass PINNs in accuracy for this benchmark, their interpretability, reduced parameter count, and theoretical alignment with classical approximation methods suggest strong potential. Future work should focus on optimizing their computational performance and exploring adaptive sampling schemes to make them competitive for large-scale shell simulations.

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